0.1 Introduction

This paper aims to forecast electricity prices in the European day-ahead market for the Germany / Luxembourg bidding zone. In this market, a daily auction is held at 12:00 CET to determine electricity prices for the following day (EPEX Spot, n.d.). To predict the outcome of this auction, machine learning models are trained and evaluated on a dataset spanning from 2018 to 2025 and containing a number of meteorological, economic, grid-specific and calendar features.

This paper uses day-ahead price data from the European Network of Transmission System Operators for Electricity (ENTSO-E) to train regression, ARIMA, random forest, and LSTM models with the aim of predicting day-ahead electricity prices, and benchmarks the performance of these models against a naive forecast.

0.2 Methodology

The price dataset was obtained from the ENTSO-E Transparency Platform through the ENTSO-E API (ENTSO-E, 2025). Since the market time unit (MTU) of the German Day-Ahead Market switched from hourly to 15 minutes on 30 September 2025, all prices were resampled into hourly prices using the mean. This approach was adopted to limit artificial data creation, but comes with the limitation that the models presented in this paper only predict hourly prices, not the 15-minute prices currently found in the day-ahead market.

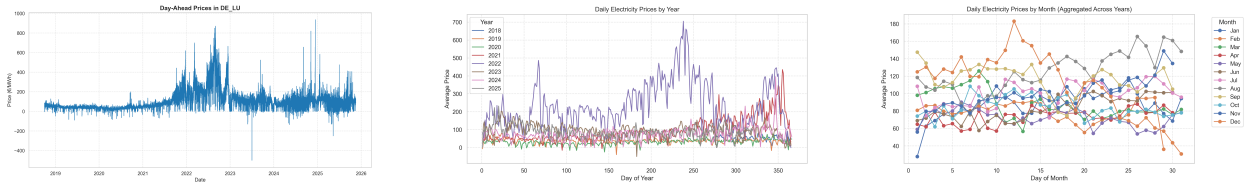
Data on electricity consumption (load) was imported from the Strommarkt-Auswertungs- und Reporting-Dashboard (SMARD) from the German Federal Network Agency (SMARD, 2025), as the load dataset from ENTSO-E only had observations for the month of November in each year. Another grid-specific feature used is generation activity, which was obtained from ENTSO-E. This dataset includes generation in MW from various sources, such as biomass, fossil brown coal / lignite, geothermal, hydro, solar and wind generation. To obtain the net contribution of hydro energy, the electricity consumed by pumped-storage hydro plants during water pumping was subtracted from the total energy produced by these plants.

Meteorological data for Germany was included due to the high share of renewable energy in Germany from solar and wind generation. Hourly temperature in Celsius, Wind Speed in km/h, cloud coverage, and precipitation in millimeter data was obtained for each Bundesland from Open Meteo through the Open Meteo API (Open-Meteo, 2025). Since the resulting dataset contained 64 columns, leading to high dimensionality and significantly slowing down the model training process, principal component analysis was used to extract the first 21 principal components of the weather data, retaining 85% of the data.

Since a substantial share of energy is not generated from renewable sources, economic variables such as the price of Brent crude oil, natural gas, and EU Allowances (EUAs) for the emission of one metric ton of CO_2. These prices were imported from Bloomberg, specifically from the CO1 Comdty, NG1 Comdty, and MO1 Comdty tickers. Additionally, to measure economic uncertainty, which may significantly influence electricity prices, data on the VDAX index, which measures implied volatility of DAX index options, as well as 10-year and 2-year German bond yields were collected from the V1X Index, GDBR10 Index, and GDBR2 Index respectively. By subtracting the 2-year bond yield from the 10-year bond yield, the bond yield difference feature was computed, which is commonly used to gain insight into the yield curve shape, acting as an indicator for economic expectations. All Bloomberg features were originally only available in daily frequency, and were subsequently upsampled to hourly frequency. Furthermore, the calendar features hour, day of week, and day of year, and prices lagged by 24 hours, 48 hours, and 168 hours were included as predictors.

After evaluating the performance of two naive forecasts, a simple linear regression model and a LASSO regression are explored. Subsequently, three ARIMA-type models are trained, and a Random Forest is explored. Finally, a Long Short-Term Memory (LSTM) model is trained to explore whether a deep learning architecture can outperform simpler models. The dataset was chronologically split into 20% testing data and 80% training data. To ensure robust model evaluation during model training and hyperparameter tuning, rolling-window cross-validation was used. The regression models were evaluated with an expanding window which appends 24h chunks of test data to the training data after evaluating the model on them. ARIMA models were evaluated with a rolling testing window consisting of 365 days of training data and 1 day of testing data to reduce training time. For the purpose of computational efficiency, the training window was shortened to 90 days to evaluate the ARIMA-type models. The Random Forest was tuned using RandomizedSearchCV to predict the 24-hour day-ahead vector. To better capture time dynamics, the features are restructured by day with hourly vectors and then predicted using a MultiOutputRegressor to model each hour independently. All model results are presented in Table 4.

0.3 Exploratory Data Analysis



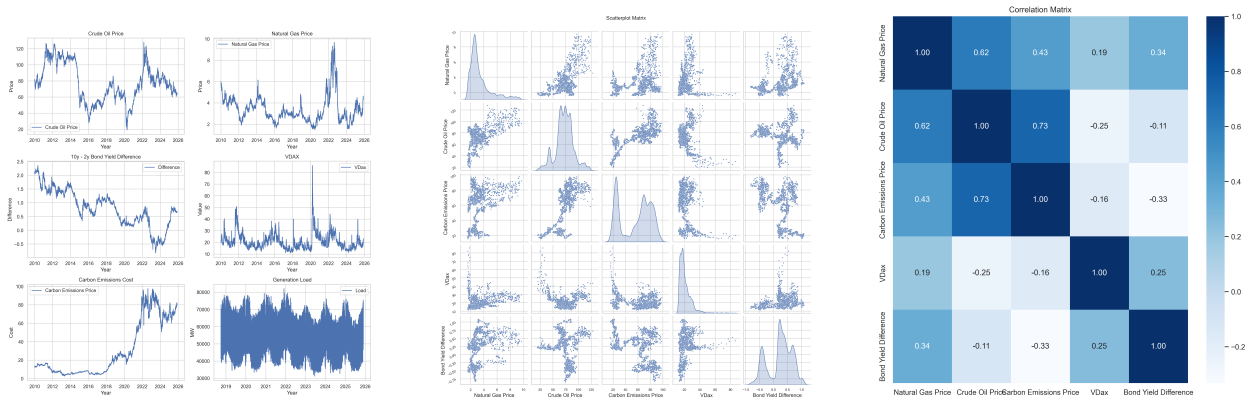
(a) Day Ahead Prices (b) Prices By Year (c) Prices By Month

Figure 1: Day-Ahead Electricity Price Trends

Figure 1 displays the yearly and monthly trends in the electricity price dataset. No clear annual or monthly trend can be spotted, and electricity prices also do not appear to rise year over year, although significant volatility can be noted in some years. Hence, seasonal and trend decomposition did not appear appropriate, and the models were directly fit to the price series. An Augmented Dickey-Fuller test on the electricity price dataset yields an ADF Statistic of -8.69 with an associated p-value of $4.07 * 10^{-14}$. This further indicates that the time series is stationary, and no unit root is present. However, an ARIMA model does perform better than a simple ARMA model, indicating that differencing may still be beneficial.

From Figure 2(a) it can be seen that a number of predictor variables, such as crude oil, natural gas, and EUA prices, may have trends or seasonality, indicating that differencing these time series may be appropriate. Figure 2(b) and 2(c) indicate there may be some correlation between predictors, especially between the prices of natural gas, crude oil, and carbon emissions (EUAs). No measures are taken at this stage to reduce the risk of multicollinearity.

Running an ADF Test on all predictor variables indicated that only the price of natural gas, crude oil, EUAs, and the bond yield difference are not stationary. Hence, these were differenced once, after which the ADF test result indicated that they were stationary.



(a) Feature Trends (b) Scatterplot Matrix (c) Correlation Matrix

Figure 2: Main Predictor Variables

0.4 Naive Forecast

In order to benchmark later models, two naive forecasts are first developed. The first naive forecast simply predicts the prices in the last 24 hours. This forecast likely performs best when electricity prices remain stable, but fails to pick up on large swings in prices. The second naive forecast predicts the average of all prices in past years at the same date and time. Hence, this naive forecast may better take into account seasonal trends.

Rolling window cross-validation with a training size of 720 hours (one month) and a testing size of 24 hours (one day). After training and testing on a fold, the window moves forward by 24 hours. This ensures a robust evaluation of the naive forecasts.

Table 1: Naive Benchmarks

Naive Forecast	**MSE**	**RMSE**	**MAE**
Last 24 hours	2069.06	37.53	28.62
Historical average	3633.01	51.70	44.57

Since the first naive forecast performs better on the testing data than the second naive forecast, the first naive forecast will serve as the benchmark against which later models are compared.

0.5 Regression Model

The full dataset with all feature variables was used to train the regression models. Additionally, lagged features and calendar features were introduced. The two lagged features are prices lagged by 24 hours and prices lagged by 168 hours. The calendar features are the hour, day of the week, and month.

First, a simple linear regression model was trained on the data. Subsequently, a LASSO regression was attempted, to see if regularization could help reduce overfitting and hence produce better predictions. This is because the L1 penalty can shrink some coefficients in the regression model exactly to zero. Rolling window cross-validation was again used to evaluate model performance, with a slightly larger training window of 8760 hours (one year), a testing window of 24 hours (one day), and a step size of 720 hours (one month). This allows the models to be trained on sufficient

data, allowing them to pick up on seasonality, but still ensures a robust model evaluation. For the LASSO regression, the train and test data was scaled within each fold using Z-score standardization, to avoid the L1 penalty unfairly shrinking large coefficients. Both the testing and training set are scaled with the mean and standard deviation of the training set to avoid data leakage. To find the optimal regularization parameter (alpha) for the LASSO regression, five different values of alpha were evaluated using rolling window cross-validation: 0.001, 0.01, 0.1, 1, and 10. The optimal alpha obtained was 1.

As the metrics in Table 4 show, both models perform better than the naive forecast, but the LASSO regression results in a slightly higher error than the ordinary linear regression without regularization. This indicates that an L1 penalty term may not help improve model performance, despite the high number of features in the dataset.

0.6 ARIMA-Type Models

Autoregressive Integrated Moving Average (ARIMA) models are commonly used in time-series forecasting and combine both autoregressive and moving average modelling (Box & Jenkins, 1976). Rolling cross validation was used to evaluate a number of different orders of p, d, and q for the ARIMA model. An optimal order of (1, 1, 3) was obtained. Subsequently, different orders of seasonal differencing were attempted for a SARIMA model, and also evaluated through rolling window cross-validation. The optimal seasonal order obtained was (1, 0, 1, 24). Finally, exogenous predictors were included as part of a SARIMAX model. All results are presented in Table 4: as shown, the SARIMA model constitutes a significant improvement over the ARIMA model, cutting the RMSE nearly in half.

0.7 Random Forest Model

A random forest model is a type of ensemble model which combines multiple decision trees trained on bootstrapped samples (Breiman, 2001). It further de-correlates trees by only considering a subset of all predictors at each split when growing the decision trees. Random Forests may work well for electricity price forecasting, especially with the dataset in this paper, as they are not affected by high-dimensional data or correlated features, and can capture non-linear relationships.

In the dataset for the random forest, two additional features were included: prices lagged by 24 hours, and prices lagged by 48 hours. A chronological 80/20 split was used for creating training and testing data.

Hyperparameter	**Values**
n_estimators	**100**, 200
max_depth	4, 6, 8, **16**, None
min_samples_leaf	4, 6, **10**
bootstrap	**True**
max_samples	0.4, 0.6, **0.8**
max_features	**sqrt**

Table 2: Random Forest Hyperparameter Grid

Table 5 summarises the different hyperparameters that were tried in search of the optimal combination for the random forest model, with the optimal hyperparameters highlighted in bold.

A randomized hyperparameter grid search was used which randomly searches 20 combinations of hyperparameters from the grid. These were then evaluated using expanding-window cross-validation.

As demonstrated in Table 4, the random forest model does not outperform the naive forecasts, the ARMA model or the SARIMA model.

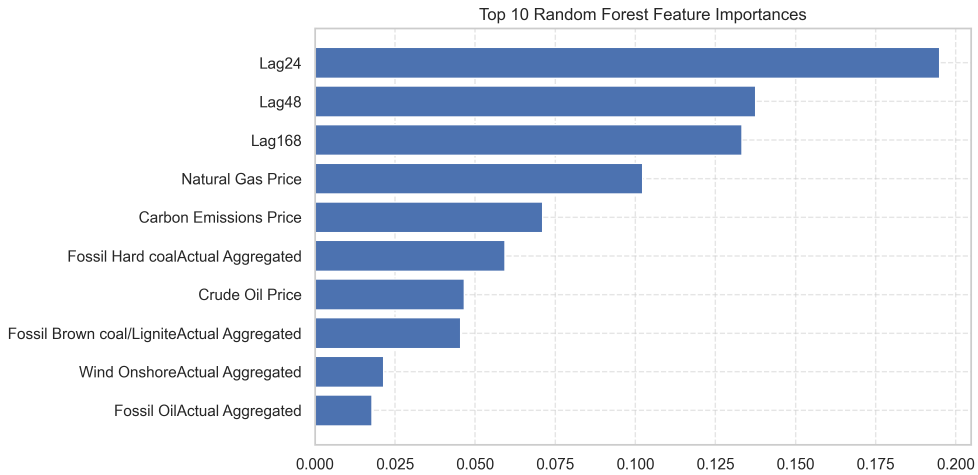


Figure 3: Random Forest Feature Importance Plot

Figure 3 shows the top ten features in the random forest model by permutation importance. It can be noted that prices lagged by 24 and 48 hours have the greatest importance, followed by the prices of natural gas, carbon emissions and crude oil. It is interesting to note that none of the weather principal components are among the top 10 features by permutation importance, despite the high share of renewable energy in Germany.

0.8 LSTM Model

Long Short-Term Memory models build upon RNN architectures by including a long-term memory state, allowing them to capture long-distance relationships in the data (Hochreiter & Schmidhuber, 1997). Due to the number of hyperparameters, this paper did a grid search to optimise the architecture of the model, exploring different number of layers, neuron per layers, adding a dropout rate to avoid overfitting and changing learning rate. The hyperparameter grid is summarised in table 6, with the optimal values obtained highlighted in bold.

Hyperparameter	**Values**
n_{units}	32, 64, 128, **256**
$layers$	1, **2**, 3
$dropout$	**0**, 0.15
$learning_rate$	0.001, **0.0005**

Table 3: LSTM Hyperparameter Grid

The results were less promising than expected, and the LSTM ended up being one of the weaker models explored in this paper. It can be speculated that the main cause is the limited amount of data used to train the model, as deep learning architectures tend to be quite data-hungry. As discussed earlier, data collection had to be stopped in 2018 due to structural changes in the market,

leaving only seven years of usable data. Moreover, many of the complex features that LSTMs are designed to leverage were not fully present in the dataset.

Given the poor performance of the initial LSTM model, this paper did not investigate further improvements in detail and instead focused on simpler, more robust models that achieved better results with the available dataset.

0.9 Discussion of Results

Table 4: Summary of Test Set Results

Metric	**Regression**	**LASSO**	**ARIMA**	**SARIMA**	**SARIMAX**	**Random Forest**	**LSTM**
MSE	1477.67	1504.22	2634.72	601.56			
RMSE	31.80	31.90	45.69	23.07			
MAE	25.19	25.25	36.16	18.52			

Some of the models attempted in this paper manage to outperform a naive forecast which simply predicts the prices observed in the last 24 hours, whereas other models perform worse than this naive forecast, and hence do not appear suitable for use in electricity price prediction. In particular, regression and LASSO regression perform quite well. The LSTM model performed poorly on the dataset provided, which may in part be attributed to the fact that deep learning architectures often require larger datasets to perform well.

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